

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

ARTIFICIAL INTELLIGENCE [IT-4401]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A warehouse robot uses a goal-based agent model. **Examine** how the environment type (stochastic vs. deterministic) influences its behavior. (5)
 - A)
 - B) **Create** a custom heuristic function for a map navigation problem using informed search. Justify your design. (5)
- 2) Assume given a two-player strategy game, **construct** the AND-OR tree and explain how it optimizes decision making. (5)
 - A)
 - B) **Analyze** a case where proof by contradiction is more efficient than direct proof in a propositional logic system. (5)
- 3) An e-commerce system uses First Order Logic to infer customer intent. **Design** how rules can support personalization. (5)
 - A)
 - B) **Compare** the challenges of reasoning with default knowledge in autonomous driving systems vs. categorization using ontologies. (5)
- 4) In a smart home system, how can Bayes' Rule be applied to determine if someone is home based on sensor data? (5)
 - A)
 - B) Given a Bayesian Network with the following structure
Cloudy $\rightarrow$ Sprinkler
Cloudy $\rightarrow$ Rain
Sprinkler $\rightarrow$ WetGrass
Rain $\rightarrow$ WetGrass
Write the joint probability distribution **P(Cloudy, Sprinkler, Rain, WetGrass)** as a product of **conditional probabilities** based on the structure of the network.
Express **P(WetGrass | Sprinkler, Rain)** in terms of **conditional probability table**. (5)
- 5) **Evaluate** a scenario where a multi-attribute utility function is applied in a self-driving (5)

car's decision-making process.

A)

B) Analyse a decision network for a robot vacuum choosing between cleaning, charging, or returning home. (5)

-----End-----